

Mock exam 1

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Does the speed with which the car is moving affect the distance needed to stop?

You work on the development of an algorithm that can identify vehicles that move at potentially dangerous speeds using data from cameras. Your part of the project is to develop a suitable statistical model. To train the algorithm, your colleagues measured the speed of various cars (in miles) and the distance they pass before they stop (in feet).

The original data can be found in the `cars` dataset (`data(cars)`).

Questions

- Load the dataset and convert columns to the metric form – from feet to meters and miles per hour to kilometers per hour. 1 meter is equal to 3.28 feet and 1 mile is equal to 1609.344 meters. (3 points.)
- Build the appropriate regression model. Check the assumptions for the use of this model. (5 points.)
- Check for any abnormal values. Justify your decision (3 points).
- If the assumptions are satisfied, discuss the model. How much variance is explained? Is the correlation statistically significant? What is the relationship between the speed and the distance to stop? (5 points.)
- Build a plot that shows all the data points and depicts the model on it. (6 points.)
- The cameras equipped with your algorithm captured 5 cars. Their speed is 35 and 68 km/h. What is the predicted distance needed to stop? Use R to make these predictions. (3 points.)

Originally created by Dmytro Shytikov in 2025.